Enhancing Video-LLM Reasoning via Agent-of-Thoughts Distillation

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Abstract

Video question answering demands multi-step reasoning and precise spatial-temporal grounding, capabilities that many Video Large Language Models (Video-LLMs) lack despite strong benchmark scores. Agent-based systems offer interpretable, tool-grounded reasoning but are slow and brittle at inference time; end-to-end Video-LLMs are efficient but opaque and error-prone on complex, compositional queries. We propose Agent-of-Thoughts Distillation (AoTD), a framework that bridges these two paradigms by distilling reliable, tool-grounded reasoning chains into a Video-LLM to jointly learn answers and rationales. AoTD builds an agent system that decomposes video queries into sub-tasks—temporal grounding, object detection, action recognition, and single-hop QA—and executes them with best-in-class specialist models, producing execution traces. These traces are converted into natural-language Chain-of-Thoughts (CoTs) and verified by an LLM for correctness and logical coherence. The filtered CoTs are then distilled into a Video-LLM via dual-prompt instruction tuning, jointly optimizing over answers and rationales. Experiments show that AoTD consistently improves LLaVA-NeXT-Video on both multiple-choice and open-ended VideoQA benchmarks (e.g., MVBench: 53.4% to 55.6%, ANetQA: 47.1% to 53.9%), while reducing inference latency from 47.93s to 10.58s and memory from 65.34GB to 18.21GB compared to the agent system. AoTD is model-agnostic and transfers effectively to Onevision, VideoLLaMA2, and Qwen2-VL.

1. Introduction

Video question answering (VideoQA) requires a model to perceive fine-grained spatial-temporal dynamics, perform multi-step compositional reasoning, and produce accurate, grounded answers. Recent Video Large Language Models such as LLaMA-VID [19], Video-LLaVA [20], VideoChat2 [17], VideoLLaMA2 [4], and LLaVA-NeXT-Video [47] have achieved impressive results by training large vision-language models end-to-end. However, these models often struggle on challenging compositional and temporal queries, generating answers in a single forward pass without any explicit step-by-step reasoning.

A complementary paradigm addresses complex visual queries through agent-based systems [7, 13, 14, 25, 34], where a controller decomposes the question into sub-tasks and invokes specialized vision tools such as object detectors, temporal grounders, and action recognizers. Such systems produce explicit execution traces that are interpretable and verifiable. However, they suffer from high inference latency due to repeated model invocations, large memory footprints from loading multiple specialist models, and brittleness to imperfect code generation.

We observe that these two paradigms are complementary: agent systems produce high-quality, step-wise reasoning traces but are slow; Video-LLMs are fast but lack explicit reasoning. This motivates a natural question: *can we distill the interpretable reasoning of an agent system into a single Video-LLM?*

We answer this affirmatively with **Agent-of-Thoughts Distillation (AoTD)**, as illustrated in Figure 1. AoTD operates in three main stages. First, an agent system decomposes each video query into sub-tasks and executes them using best-performing specialist models, collecting intermediate results and execution traces. Second, these program traces are converted into fluent natural-language CoTs via an LLM-based transfer step, then filtered by an LLM verifier to retain only correct and coherent rationales. Third, the verified CoTs are used to instruction-tune a Video-LLM using dual prompts—one requesting only the final answer and one requesting step-by-step reasoning—jointly optimizing both objectives. The full pipeline is depicted in Figure 2.

Our main contributions are:

- We propose **AoTD**, a distillation framework that imbues Video-LLMs with multi-step spatial-temporal reasoning by learning from agent-generated, LLM-verified CoTs.
- We construct an agent system that selects top-performing specialist models per sub-task via systematic evaluation, and converts execution traces into natural-language CoTs.
- We curate and filter a dataset of **32.3K high-quality CoTs** across compositional and temporal VideoQA

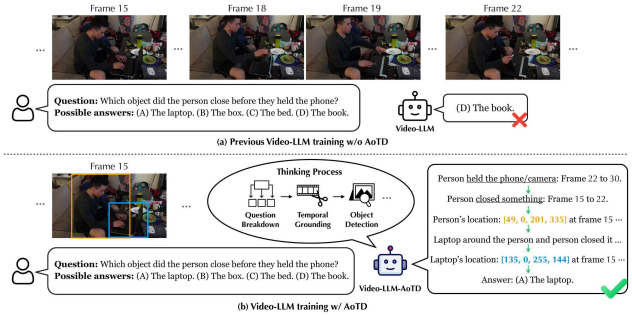

Figure 1. **Overview of AoTD.** Our method distills multi-step reasoning and spatial-temporal understanding into a single generative Video-LLM, achieving both the interpretability of agent systems and the efficiency of end-to-end models.

benchmarks for instruction tuning.

- We demonstrate consistent gains over strong baselines on MC-VQA and OE-VQA benchmarks, with emergent spatial-temporal grounding and markedly reduced inference cost versus the agent system.

2. Related Work

Video Large Language Models. Building on large vision-language models [1, 16, 22], a growing body of work extends them to video. Video-LLaVA [20] aligns video and language in a shared feature space. LLaMA-VID [19] represents each frame with two tokens for efficient long-video processing. VideoChat2 [17] introduces instruction tuning on large-scale video data and the MVBench suite. VideoLLaMA2 [4] adopts a spatiotemporal convolution connector for richer temporal encoding. LLaVA-NeXT-Video [47] extends the LLaVA-NeXT architecture to video by sampling frames uniformly. Despite their advances, these models generate answers in a single pass and lack explicit compositional reasoning and spatial-temporal grounding capabilities.

Agent-based Visual Question Answering. Agent systems tackle complex visual queries by decomposing them into sub-tasks and invoking specialized tools. ViperGPT [34] uses GPT-4 to generate Python programs that call vision APIs for image QA. VideoAgent [7] adapts this paradigm to video by iteratively querying relevant frames. AVIS [14] learns an adaptive visual information-seeking strategy. VPD [13] focuses on visual program distillation for images. MoreVQA [25] decomposes video queries modularly. DoraemonGPT [41] builds a comprehensive video agent with memory and planning. ProViQ [5] generates programs for video QA. While interpretable, these systems incur heavy inference costs from sequential model calls, which AoTD addresses by distilling their reasoning into a single efficient model.

Chain-of-Thought Reasoning for VLMs. Chain-of-Thought (CoT) prompting [36] has proven effective for multi-step reasoning in language models. VisCOT [33] applies visual CoT to vision-language models for grounded reasoning. CCOT [28] generates compositional CoTs for image understanding. Orca [27] distills step-by-step reasoning traces from GPT-4 into smaller models. mCoT [48] explores multimodal CoT for science QA. VideoThought [8] explores reasoning chains for video understanding. VSI-Bench [40] evaluates spatial reasoning in video. Unlike prior work, AoTD generates CoTs from agent execution traces for video, providing spatial-temporal grounding that is verified before distillation.

Knowledge Distillation for Vision-Language Models. Knowledge distillation transfers capabilities from large teacher models into compact students [**?**]. For language models, Orca [27] distills explanation traces from GPT-4 to improve smaller-model reasoning without requiring proprietary inference at test time. For VLMs, prior efforts focus on distilling image recognition or captioning capabilities. AoTD takes a distinct approach: the teacher is a heterogeneous multi-model agent system rather than a single larger network, and the distilled signal consists of grounded tool-execution traces that encode spatial-temporal structure unavailable from any single generative teacher model alone. This design is especially suited to VideoQA, where specialist perception models substantially outperform generalist VLMs on sub-tasks such as temporal grounding and object localization.

Compositional and Temporal VideoQA. Several benchmarks probe compositional and temporal reasoning. STAR [38] focuses on situated reasoning with four types. NExT-QA [39] tests causal and temporal question answering. AGQA [12] probes compositional spatial-temporal relations. ANetQA [45] evaluates diverse question types over ActivityNet videos. CLEVRER [42] tests spatial-temporal causal reasoning. EgoQA [11] covers egocentric video. VideoStar [49] further explores spatio-temporal reasoning. AoTD targets these challenging benchmarks by instilling explicit compositional reasoning into Video-LLMs.

3. Agent-of-Thoughts Distillation

Figure 2 provides a full overview of AoTD, comprising (1) agent-based CoT construction, (2) CoT transfer from traces to natural language, (3) LLM-based filtering, and (4) dual-prompt distillation.

3.1. Agent-based CoT Construction

Sub-task Decomposition. Given a video V and a query Q (with possible answers $\mathcal{A}$ for MC settings), the agent system

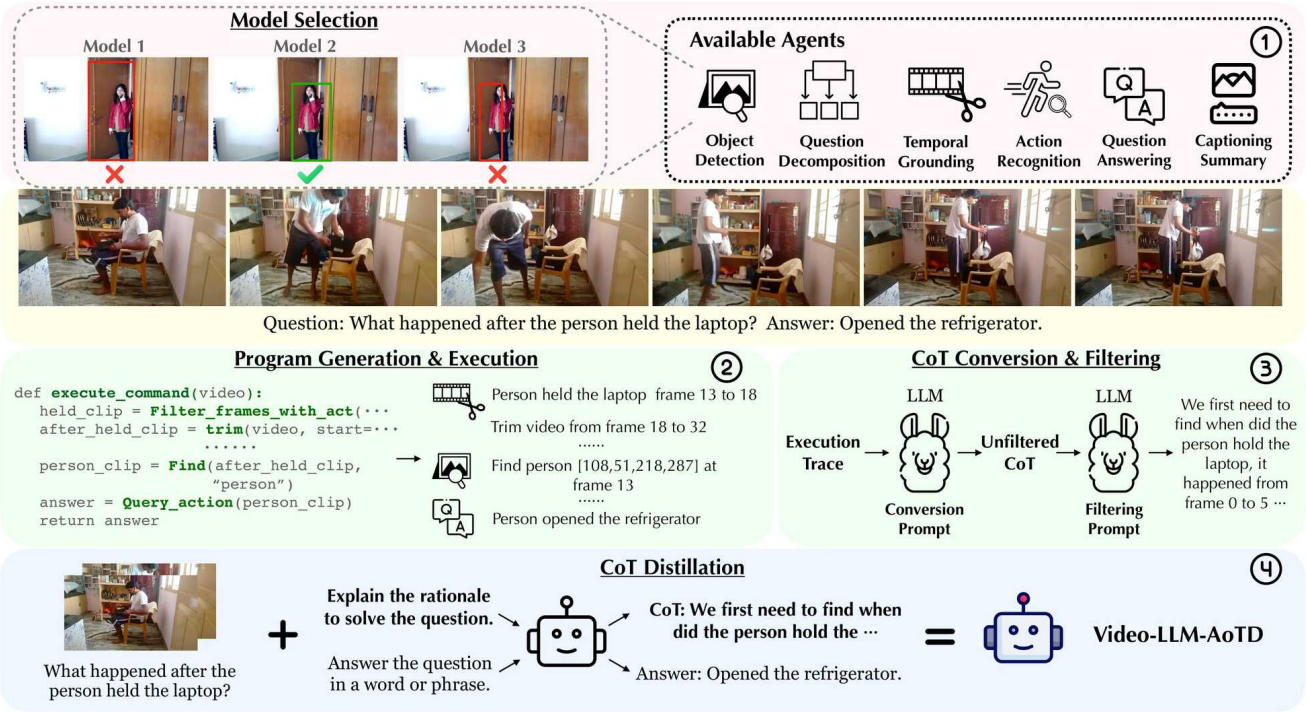

Figure 2. **AoTD pipeline.** Given a video and query, the agent system decomposes the query into sub-tasks, executes them with specialist models, and collects execution traces. An LLM converts these traces into natural-language CoTs, which are then verified and used to instruction-tune a Video-LLM via dual-prompt distillation.

uses a code-generation LLM to produce a Python program that decomposes Q into a sequence of specialized sub-tasks. We define five sub-task types (Table 1):

1. **Question decomposition:** Parse the complex query into simpler sub-questions to be answered sequentially.
2. **Object detection:** Localize and identify objects in relevant video frames.
3. **Temporal grounding:** Identify the temporal segment relevant to a sub-question.
4. **Action recognition:** Classify the action or activity in a given clip.
5. **Question answering:** Answer a simple single-hop sub-question given visual or temporal context.

Specialist Model Selection. For each sub-task type, we evaluate multiple candidate models on the STAR training set and select the best performer (Table 1):

- *Question decomposition:* DeepSeek-Coder-Instruct (6.7B) [6] achieves 85.7% accuracy, outperforming CodeQwen1.5-Chat (52.7%) and GPT-3.5-Turbo (73.1%).
- *Object detection:* OWL-ViT v2 [26] achieves 63.0 IoU, outperforming OWL-ViT v1 (47.3) and GLIP [18] (58.9).
- *Temporal grounding:* UniVTG [21] achieves 24.7/35.3 IoU/Recall, outperforming LITA-13B [15] (11.7/20.2) and TimeChat-7B [32] (13.9/23.1).
- *Action recognition:* LLaVA-NeXT-Video-DPO (7B) [47] achieves 18.2% Top-1 accuracy, outperforming InternVideo2 [35] (7.6%) and Open-VCLIP [37] (8.9%).
- *Question answering:* LLaVA-NeXT-Video-DPO (7B) [47] achieves 53.4% accuracy, outperforming LLaMA-VID [19] (43.5%) and SeViLA [43] (46.5%).

Execution Trace Collection. The agent executes the generated program step-by-step, invoking selected specialist models and recording intermediate results. As shown in Figure 3, each execution produces a trace comprising (i) the generated program, (ii) intermediate outputs at each step (e.g., detected objects, temporal segments, sub-answers), and (iii) the final predicted answer. Only traces where the predicted answer matches the ground-truth label are retained.

3.2. CoT Transfer

Raw execution traces consist of program code and structured intermediate outputs, unsuitable for direct instruction tuning. We convert them into fluent natural-language rationales using LLaMA-3.1-8B-Instruct with a dedicated transfer prompt. This prompt instructs the LLM to narrate the reasoning steps coherently, describing what each sub-task

Sub-task	Model name	Metric	Number (%)
question decomposition	CodeQwen1.5-Chat (7B) [2]	Acc	52.7
	GPT-3.5-Turbo [29]		73.1
	DeepSeek-Coder-Instruct (6.7B) [6]		85.7
object detection	OWL-ViT v1 [24]	IoU	47.3
	GLIP [18]		58.9
	OWL-ViT v2 [26]		63.0
temporal grounding	LITA (13B) [15]	IoU / Recall	11.7 / 20.2
	TimeChat (7B) [32]		13.9 / 23.1
	UniVTG [21]		24.7 / 35.3
action recognition	InternVideo2 (1B) [35]	Top1-Acc	7.6
	Open-VCLIP [37]		8.9
	LLaVA-NeXT-Video-DPO (7B) [47]		18.2
question answering	LLaMA-VID (7B) [19]	Acc	43.5
	SeViLA [43]		46.5
	LLaVA-NeXT-Video-DPO (7B) [47]		53.4

Table 1. Sub-tasks definition and evaluation results. We choose 3 model candidates for each sub-task and evaluate them in STAR training set with the corresponding metrics. Models with best performance are placed at the bottom of each column.

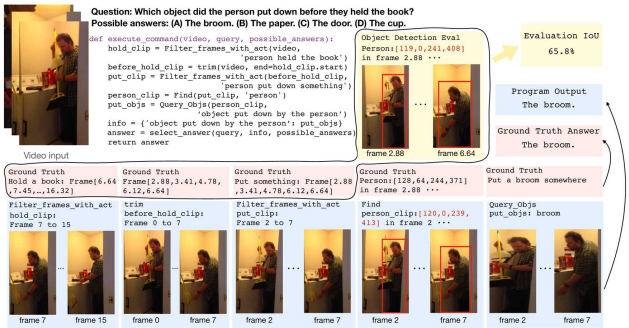

Figure 3. **Execution process.** An example of step-by-step program execution within the agent system, showing sub-task decomposition, specialist tool invocation, and intermediate result collection.

found and how intermediate results led to the final answer. The output is a natural-language CoT such as: "*First, I identify the temporal segment in which the action occurs... Then, I detect the objects present... Based on these observations, the answer is...*"

3.3. CoT Filtering

Not all transferred CoTs are logically coherent or faithfully reflect the video content. We apply LLM-based verification using LLaMA-3.1-8B-Instruct with a filtering prompt: given the question and candidate rationale, the LLM outputs `True` if the rationale is logically consistent and correctly leads to the given answer, or `False` otherwise. Only CoTs passing this filter are retained for training. Table 6 shows that filtering significantly improves performance (e.g., MVBench: 53.7% to 55.6%, STAR: 73.3% to 74.3%).

The filtering prompt supplies the LLM with the original query, the candidate CoT, and the ground-truth answer, asking it to verify three criteria: (i) *factual consistency*—each sub-task output is plausible given the query; (ii) *logical coherence*—intermediate results are correctly chained toward a conclusion; and (iii) *answer alignment*—the stated conclusion matches the ground-truth label. A CoT is retained only when all three criteria are met, ensuring that distilled rationales encode reliable spatial-temporal evidence rather than spurious or hallucinated reasoning steps.

3.4. Dual-Prompt Distillation

The filtered CoTs are used to instruction-tune a Video-LLM via dual-prompt training, inspired by knowledge distillation [27]. Each training sample appears in two variants:

- **Answer prompt:** The question and options with the suffix "*Answer with the option's letter from the given choices directly.*" Target: the ground-truth answer letter.
- **Rationale prompt:** The same question and options with the suffix "*Explain the rationale to answer the question.*" Target: the filtered CoT.

Both variants are included in each training batch. The model is fine-tuned with standard cross-entropy loss over target tokens, jointly optimizing answer prediction and rationale generation. This dual objective encourages the model to internalize step-wise spatial-temporal reasoning while retaining concise-answer capability. We fine-tune LLaVA-NeXT-Video-7B [47] with a maximum sequence length of 8192 tokens and a batch size of 4.

3.5. Training Data

We construct CoTs for six VideoQA datasets covering complementary reasoning skills (Table 2). For MC benchmarks: STAR [38] (compositional, 45.7K labels, 11.2K CoTs), NExT-QA [39] (temporal and causal, 34.1K labels, 12.1K CoTs), CLEVRER [42] (spatial-temporal, 21.0K labels). For OE benchmarks: AGQA [12] (compositional, 25.0K labels, 5.4K CoTs), ANetQA [45] (compositional, 25.0K labels, 3.6K CoTs), and EgoQA [11] (egocentric, 7.8K labels). In total we process 158.6K VideoQA pairs and generate **32.3K high-quality CoTs** after filtering. The per-dataset retention rates vary considerably: STAR retains 24.5% of executed traces, NExT-QA 35.5%, and ANetQA 14.4%, reflecting that compositional multi-hop queries tend to produce more logically coherent rationales than temporally complex open-ended questions where the agent's sub-task outputs are noisier.

4. Experiments

4.1. Experimental Setup

Base Model and Baselines. Our primary model is LLaVA-NeXT-Video-7B [47], fine-tuned with AoTD to

Dataset	Description	# Labels	# CoTs
AGQA	Compositional	25.0K	5.4K
ANetQA	Compositional	25.0K	3.6K
STAR	Compositional	45.7K	11.2K
NExT-QA	Temporal & Causal	34.1K	12.1K
CLEVRER	Spatial & Temporal	21.0K	-
EgoQA	Ego-centric	7.8K	-
Total		**158.6K**	**32.3K**

Table 2. Dataset statistics. The column "# Labels" indicates the number of VideoQA pairs, which include the video, query, possible answers (multiple-choice), and the correct answer. "# CoTs" refers to the number of CoTs generated using our agent-based system for each dataset.

obtain **LLaVA-NeXT-Video-AoTD** (LNV-AoTD). The main baseline is **LLaVA-NeXT-Video-Instruct** (LNV-Instruct), instruction-tuned on the same datasets but without CoT rationales. We also compare against VideoLLaMA [46], Video-ChatGPT [23], LLaMA-VID [19], Video-LLaVA [20], VideoChat2 [17], VideoLLaMA2 [4], and proprietary models Gemini 1.5 Pro [10] and GPT-4V [30].

Evaluation Benchmarks. We evaluate on the datasets in Table 3. For MC-VQA: MVBench [17], VideoMME [9], STAR [38], NExT-QA [39], Perception-Test [31], and VSI-Bench [40]. For OE-VQA: ANetQA [45], AGQA [12], ActivityNet-QA [44], and Video-ChatGPT [23].

Implementation Details. We uniformly sample 32 frames per video. We use the AdamW optimizer with a peak learning rate of 2e-5, weight decay of 0.05, and linear warmup over 3% of total steps followed by cosine decay. Training runs for 1 epoch on 8 NVIDIA A100 (80GB) GPUs with gradient checkpointing enabled to handle the extended sequence lengths introduced by CoT rationales. Both answer and rationale prompt variants are included in every training batch.

4.2. Main Results on MC-VQA

Table 4 reports results on six MC-VQA benchmarks. LNV-AoTD achieves consistent gains over LNV-Instruct across all benchmarks: MVBench (+2.2%), VideoMME (+1.8%), STAR (+2.1%), NExT-QA (+0.5%), Perception-Test (+0.3%), and VSIBench (+2.1%). Compared to other 7B open-source models, LNV-AoTD outperforms VideoLLaMA2 on MVBench (53.4 vs. 55.6) and VideoChat2 (51.1 vs. 55.6) while using the same backbone architecture.

4.3. Main Results on OE-VQA

Table 5 shows results on OE-VQA benchmarks. LNV-AoTD achieves substantial gains over LNV-Instruct on ANetQA (47.1/3.1 to 53.9/3.4 Acc/Score) and ActivityNet-QA (50.0/3.3 to 53.2/3.4). On AGQA, accuracy improves from 59.3 to 60.9. Video-ChatGPT metrics similarly improve: Correctness (2.96 to 3.11), Detail (2.81 to 3.00), and Context (3.35 to 3.60), demonstrating that distilled rationales enrich the model's descriptive capabilities.

Dataset	Size		Type	Train	Eval
	train	eval			
MC-VQA					
STAR [38]	45.7K	7.1K	Compositional	✓	✓
NExT-QA [39]	34.1K	5.0K	Temporal & Causal	✓	✓
CLEVRER [42]	21.0K	-	Spatial-temporal	✓	✗
Perception-Test [31]	-	11.5K	General	✗	✓
MVBench [17]	-	4.0K	General	✗	✓
VideoMME [9]	-	2.7K	General	✗	✓
VSIBench [40]	-	5.0K	spatial-temporal	✗	✓
OE-VQA					
AGQA [12]	25.0K	2.0K	Compositional	✓	✓
ANetQA [45]	25.0K	2.0K	Compositional	✓	✓
EgoQA [11]	7.8K	-	Ego-centric	✓	✗
Activitynet-QA [44]	-	8.0K	General	✗	✓
Video-ChatGPT [23]	-	3.0K	General	✗	✓

Table 3. Training and evaluation datasets statics.

4.4. Ablation Studies

Effect of CoT Filtering. Table 6 (upper section) compares LNV-AoTD with and without filtering. Removing filtering degrades performance on all benchmarks: MVBench drops from 55.6% to 53.7%, STAR from 74.3% to 73.3%, and AGQA from 60.9/3.6 to 59.5/3.5 (Acc/Score). This confirms that noisy or incoherent CoTs are harmful, and LLM-based verification is essential.

Effect of Dual-Prompt Training. We ablate the two training objectives in the dual-prompt strategy by training with only the answer prompt (Answer-Only) or only the rationale prompt (Rationale-Only) and comparing to the full dual-prompt AoTD. Answer-Only fine-tuning, which omits rationale supervision, yields MVBench 54.1% and STAR 72.6%—below LNV-AoTD (55.6% / 74.3%), confirming that CoT supervision contributes beyond data augmentation. Rationale-Only training, which never explicitly trains the concise-answer path, achieves MVBench 54.8% and STAR 73.1%, slightly worse on OE-VQA due to format mismatch. The full dual-prompt objective consistently outperforms both ablations, demonstrating that jointly optimizing answer accuracy and rationale generation produces a synergistic benefit: rationale training regularizes the answer head toward structured reasoning, while answer training prevents the model from producing verbose outputs at the expense of correctness.

Model	MVBench (Acc.)	VideoMME (Acc.)	STAR (Acc.)	NExT-QA (Acc.)	Perception-Test (Acc.)	VSIBench (Acc.)
Proprietary Models						
Gemini 1.5 Pro [10]	-	75.0	-	-	-	45.4
GPT4-V [30]	43.7	59.9	-	-	-	-
Open-source Models						
LLaMA-VID (7B) [19]	41.9	25.9	-	-	44.6	-
Video-LLaVA (7B) [20]	41.0	39.9	-	-	44.3	-
VideoChat2 (7B) [17]	51.1	33.7	59.0*	68.6*	47.3	-
VideoLLaMA2 (7B) [4]	53.4	45.1	58.5*	62.3*	49.6	-
LLaVA-NeXT-Video (7B) [47]	46.5*	41.0*	52.4*	61.6*	47.5*	19.7*
LLaVA-NeXT-Video-Instruct (7B)	53.4	43.2	72.2	77.1	50.3	26.7
LLaVA-NeXT-Video-AoTD (7B)	55.6	45.0	74.3	77.6	50.6	28.8

Table 4. Comparison with Video-LLMs on MC-VQA benchmarks. LLaVA-NeXT-Video-AoTD outperforms all other baselines the and the version without CoT distillation. * means results reproduced by ourseleves. Results without signs are retrieved from [4] and [40].

Model	ANetQA (Acc./Score)	AGQA (Acc./Score)	Video-ChatGPT (Score)					ActivityNet (Acc./Score)
			Corr.	Deta.	Cont.	Temp.	Cons.	
Proprietary Models								
Gemini 1.5 Pro [10]	-	-	-	-	-	-	-	56.7/-
GPT4-V [30]	-	-	4.09	3.88	4.37	3.94	4.02	59.5/-
Open-Source Models								
VideoLLaMA (7B) [46]	-	-	1.96	2.18	2.16	1.82	1.79	12.4/1.1
Video-ChatGPT (7B) [23]	-	-	2.50	2.57	2.69	2.16	2.20	35.2/2.7
LLaMA-VID (7B) [19]	-	-	2.96	3.00	3.53	2.46	2.51	47.4/3.3
Video-LLaVA (7B) [20]	-	-	2.87	2.94	3.44	2.45	2.51	45.3/3.3
VideoChat2 (7B) [17]	-	-	3.02	2.88	3.51	2.66	2.81	49.1/3.3
VideoLLaMA2 (7B) [4]	-	-	3.09	3.09	3.68	2.63	3.25	49.9/3.3
LLaVA-NeXT-Video (7B) [47]	46.4/3.3*	27.4/2.2*	3.26*	3.22*	3.77*	2.47*	2.99*	54.3/3.2*
LLaVA-NeXT-Video-Instruct (7B)	47.1/3.1	59.3/3.4	2.96	2.81	3.35	2.42	2.82	50.0/3.3
LLaVA-NeXT-Video-AoTD (7B)	53.9/3.4	60.9/3.6	3.11	3.00	3.60	2.41	2.91	53.2/3.4

Table 5. Comparison with Video-LLMs on OE-VQA benchmarks. LLaVA-NeXT-Video-AoTD improves performance in all open-ended benchmarks compared with the Instruct version. * means results reproduced by ourseleves. Results without signs are retrieved from [4].

Effect of CoT Data Scale. Training LNV-AoTD with 25%, 50%, and 100% of the 32.3K CoTs yields MVBench accuracies of 54.2%, 55.1%, and 55.6%, respectively. This monotonic improvement indicates that performance has not yet saturated, and collecting additional agent-verified CoTs remains a promising avenue for further gains.

Transferability to Other Video-LLMs. Table 6 (lower section) shows that our CoT dataset transfers to other Video-LLMs. Applying AoTD CoTs to Onevision [3] yields MVBench 59.2 to 60.5 and STAR 75.8 to 76.6. VideoLLaMA2 [4] similarly benefits (MVBench 54.9 to 56.0, STAR 69.2 to 71.1). Qwen2-VL improves on all three metrics. These results confirm that AoTD is model-agnostic.

Model	Filtering	MVBench (Acc.)	STAR (Acc.)	AGQA (Acc. / Score)
LNV-AoTD	✗	53.7	73.3	59.5/3.5
LNV-AoTD	✓	55.6	74.3	60.9/3.6
Onevision-Instruct	-	59.2	75.8	65.6/3.7
Onevision-AoTD	✓	60.5	76.6	65.7/3.7
VideoLLaMA2-Instruct	-	54.9	69.2	56.0/3.5
VideoLLaMA2-AoTD	✓	56.0	71.1	57.2/3.5
Qwen2-VL-Instruct	-	65.6	71.4	59.8/3.6
Qwen2-VL-AoTD	✓	66.5	73.1	61.2/3.7

Table 6. Ablation results of CoT filtering and transferability.

4.5. Efficiency Analysis

Table 7 compares the agent-based system and LNV-AoTD at inference time. The agent system requires 47.93s per sample (sequential model invocations), 65.34GB memory (multiple specialist models loaded simultaneously), and 30.98 TFLOPs. LNV-AoTD reduces latency to 10.58s (4.5× speedup), memory to 18.21GB (3.6× reduction), and computation to 13.53 TFLOPs (2.3× reduction). AoTD retains the reasoning quality of the agent system while achieving single-model efficiency. This efficiency gain is espe-

cially important for latency-sensitive applications such as video surveillance, autonomous driving assistance, or interactive video search, where the agent system's 48s-per-query cost would be entirely impractical for deployment.

Model	Time (s) ↓	Memory (GB) ↓	TFLOPs ↓
Agent-based system	47.93	65.34	30.98
LNV-AoTD	10.58	18.21	13.53

Table 7. Latency and computation comparison.

4.6. Spatial-Temporal Grounding Analysis

Table 8 evaluates grounding capabilities. LNV-Instruct cannot produce temporal or spatial predictions (indicated by ✗), as it lacks grounding supervision. In contrast, LNV-AoTD achieves temporal grounding IoU of 21.7% and Recall of 34.0%, comparable to the specialist UniVTG [21] (22.8 IoU / 31.0 Recall). For spatial grounding, LNV-AoTD achieves 45.2% IoU, below the specialist OWL-ViT v2 [26] (64.7%) but notable as an emergent capability from CoT distillation without explicit grounding supervision.

This emergent grounding behavior arises because rationale targets describe temporal and spatial observations in natural language, and the model learns to associate these descriptions with its internal visual representations during training. The grounding capacity emerges as a by-product of learning structured rationales rather than from dedicated localization supervision, suggesting that language-grounded CoTs implicitly encode perceptual structure that generalizes beyond QA to localization tasks.

4.7. Qualitative Analysis

Figure 4 presents qualitative examples of rationales generated by LNV-AoTD. The model correctly identifies relevant temporal segments, grounds objects in video frames, and chains these observations into coherent reasoning paths leading to correct answers. LNV-Instruct, without rationale training, directly produces answers and often fails on queries requiring multi-step spatial-temporal understanding. Importantly, errors made by LNV-AoTD tend to occur at specific reasoning steps—such as misidentifying a temporal boundary or an occluded object—rather than from a complete failure of structured reasoning. This makes failure modes transparent and diagnosable, a practical advantage over opaque single-pass predictions. The distilled rationales thus serve a dual purpose: improving accuracy and providing interpretable justifications that could support human oversight in deployment.

Error Analysis. We further categorize failure modes of LNV-AoTD to characterize its remaining limitations. Errors arise primarily in two scenarios: (i) queries requiring fine-grained counting or precise numerical reasoning, where the object-detection specialist produces imprecise localizations that propagate into the CoT and mislead subsequent reasoning steps; and (ii) questions about rare or domain-specific actions under-represented in training data, where the action-recognition specialist outputs low-confidence predictions that the CoT transfer step struggles to articulate coherently. Notably, LNV-AoTD tends to hedge rather than hallucinate confidently in these cases, suggesting that rationale supervision instills calibrated uncertainty alongside accuracy gains—a desirable property for safety-critical video understanding applications.

Model	Temporal Grounding IoU (%)	Temporal Grounding Recall (%)	Spatial Grounding IoU (%)
UniVTG	22.8	31.0	-
OWL-ViT v2	-	-	64.7
LNV-Instruct	✗	✗	✗
LNV-AoTD	21.7	34.0	45.2

Table 8. Temporal and spatial abilities evaluation results.

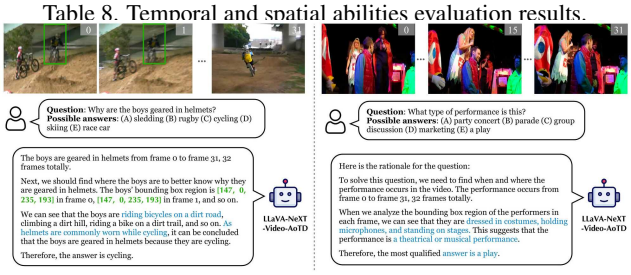

Figure 4. **Visualization of rationales.** Examples of step-by-step rationales generated by LNV-AoTD, demonstrating grounded temporal and spatial reasoning on VideoQA queries.

5. Conclusion

We presented **Agent-of-Thoughts Distillation (AoTD)**, a framework for enhancing Video-LLMs with multi-step spatial-temporal reasoning by distilling agent-generated Chain-of-Thoughts. AoTD constructs an agent system that decomposes video queries, invokes best-in-class specialist models, converts execution traces into natural-language rationales, verifies them with an LLM, and instruction-tunes a Video-LLM with dual answer-and-rationale prompts. Extensive experiments demonstrate consistent improvements over strong baselines on both MC-VQA and OE-VQA benchmarks, emergent spatial-temporal grounding capabilities, and dramatically reduced inference cost compared to agent systems. AoTD is model-agnostic and transfers effectively to Onevision, VideoLLaMA2, and Qwen2-VL, demonstrating the broad applicability of tool-grounded CoT distillation for video understanding. Current limitations include reliance on a capable code-generation LLM for subtask decomposition and higher training complexity relative to standard instruction tuning. Furthermore, the CoT filtering step discards approximately 80% of generated rationales overall, indicating that better trace-to-language con-

version methods could substantially expand the training corpus and unlock additional performance gains without requiring new video annotations. Future work will explore extending AoTD to longer videos and egocentric scenarios, incorporating retrieval-augmented sub-task execution, and designing self-improvement loops in which the Video-LLM iteratively refines its own CoT quality without requiring the full agent system at each iteration.

References

[1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. 2022. 2

[2] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, Binyuan Hui, Luo Ji, Mei Li, Junyang Lin, Runji Lin, Dayiheng Liu, Gao Liu, Chengqiang Lu, Keming Lu, Jianxin Ma, Rui Men, Xingzhang Ren, Xuancheng Ren, Chuanqi Tan, Sinan Tan, Jianhong Tu, Peng Wang, Shijie Wang, Wei Wang, Shengguang Wu, Benfeng Xu, Jin Xu, An Yang, Hao Yang, Jian Yang, Shusheng Yang, Yang Yao, Bowen Yu, Hongyi Yuan, Zheng Yuan, Jianwei Zhang, Xingxuan Zhang, Yichang Zhang, Zhenru Zhang, Chang Zhou, Jingren Zhou, Xiaohuan Zhou, and Tianhang Zhu. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023. 4

[3] Li Bo, Zhang Yuanhan, Guo Dong, Zhang Renrui, Li Feng, Zhang Hao, Zhang Kaichen, Li Yanwei, Liu Ziwei, and Li Chunyuan. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024. 6

[4] Zesen Cheng, Sicong Leng, Hang Zhang, Yifei Xin, Xin Li, Guanzheng Chen, Yongxin Zhu, Wenqi Zhang, Ziyang Luo, Deli Zhao, and Lidong Bing. Videollama 2: Advancing spatial-temporal modeling and audio understanding in video-llms. *arXiv preprint arXiv:2406.07476*, 2024. 1, 2, 5, 6

[5] Rohan Choudhury, Koichiro Niinuma, Kris M. Kitani, and Laszlo A. Jeni. Zero-shot video question answering with procedural programs. *arXiv preprint arXiv:2312.00937*, 2023. 2

[6] Guo Daya, Zhu Qihao, Yang Dejian, Dong Zhenda Xie, Kai, Zhang Wentao, Chen Guanting, Bi Xiao, Y. Wu, Y.K. Li, Luo Fuli, and Liang Yingfei, Xiongand Wenfeng. Deepseek-coder: When the large language model meets programming – the rise of code intelligence. *arXiv preprint arXiv:2401.14196*, 2024. 3, 4

[7] Yue Fan, Xiaojian Ma, Rujie Wu, Yuntao Du, Jiaqi Li, Zhi Gao, and Qing Li. Videoagent: A memory-augmented multimodal agent for video understanding. *arXiv preprint arXiv:2403.11481*, 2024. 1, 2

[8] Hao Fei, Shengqiong Wu, Wei Ji, Hanwang Zhang, Meishan Zhang, Mong-Li Lee, and Wynne Hsu. Video-of-thought: Step-by-step video reasoning from perception to cognition. 2024. 2

[9] Chaoyou Fu, Yuhan Dai, Yondong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075*, 2024. 5

[10] Gemini Team Google. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024. 5, 6

[11] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, et al. Ego4d: Around the world in 3,000 hours of egocentric video. 2022. 2, 4, 5

[12] Madeleine Grunde-McLaughlin, Ranjay Krishna, and Maneesh Agrawala. Agqa: A benchmark for compositional spatio-temporal reasoning. 2021. 2, 4, 5

[13] Yushi Hu, Otilia Stretcu, Chun-Ta Lu, Krishnamurthy Viswanathan, Kenji Hata, Enming Luo, Ranjay Krishna, and Ariel Fuxman. Visual program distillation: Distilling tools and programmatic reasoning into vision-language models. 2024. 1, 2

[14] Ziniu Hu, Ahmet Iscen, Chen Sun, Kai-Wei Chang, Yizhou Sun, David Ross, Cordelia Schmid, and Alireza Fathi. Avis: Autonomous visual information seeking with large language model agent. 2024. 1, 2

[15] De-An Huang, Shijia Liao, Subhashree Radhakrishnan, Hongxu Yin, Pavlo Molchanov, Zhiding Yu, and Jan Kautz. Lita: Language instructed temporal-localization assistant. 2024. 3, 4

[16] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: bootstrapping language-image pre-training with frozen image encoders and large language models. 2023. 2

[17] Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. Mvbench: A comprehensive multi-modal video understanding benchmark. 2024. 1, 2, 5, 6

[18] Liunian Harold Li*, Pengchuan Zhang*, Haotian Zhang*, Jianwei Yang, Chunyuan Li, Yiwu Zhong, Lijuan Wang, Lu Yuan, Lei Zhang, Jenq-Neng Hwang, Kai-Wei Chang, and Jianfeng Gao. Grounded language-image pre-training. 2022. 3, 4

[19] Yanwei Li, Chengyao Wang, and Jiaya Jia. Llama-vid: An image is worth 2 tokens in large language models. 2024. 1, 2, 3, 4, 5, 6

[20] Bin Lin, Bin Zhu, Yang Ye, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. 2024. 1, 2, 5, 6

[21] Kevin Qinghong Lin, Pengchuan Zhang, Joya Chen, Shraman Pramanick, Difei Gao, Alex Jinpeng Wang, Rui Yan, and Mike Zheng Shou. Univtg: Towards unified video-language temporal grounding. 2023. 3, 4, 7

[22] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. 2024. 2

[23] Muhammad Maaz, Hanoona Rasheed, Salman Khan, and Fahad Shahbaz Khan. Video-chatgpt: Towards detailed video understanding via large vision and language models. 2024. 5, 6

[24] Minderer Matthias, Gritsenko Alexey, Stone Austin, Neumann Maxim, Weissenborn Dirk, Dosovitskiy Alexey, Mahendran Aravindh, Arnab Anurag, Dehghani Mostafa, Shen Zhuoran, Wang Xiao, Zhai Xiaohua, Kipf Thomas, and Houlsby Neil. Simple open-vocabulary object detection with vision transformers. 2022. 4

[25] Juhong Min, Shyamal Buch, Arsha Nagrani, Minsu Cho, and Cordelia Schmid. Morevqa: Exploring modular reasoning models for video question answering. 2024. 1, 2

[26] Matthias Minderer, Alexey Gritsenko, and Neil Houlsby. Scaling open-vocabulary object detection. 2024. 3, 4, 7

[27] Arindam Mitra, Luciano Del Corro, Shweti Mahajan, Andres Codas, Clarisse Simoes, Sahaj Agarwal, Xuxi Chen, Anastasia Razdaibiedina, Erik Jones, Kriti Aggarwal, et al. Orca 2: Teaching small language models how to reason. *arXiv preprint arXiv:2311.11045*, 2023. 2, 4

[28] Chancharik Mitra, Brandon Huang, Trevor Darrell, and Roei Herzig. Compositional chain-of-thought prompting for large multimodal models. 2024. 2

[29] OpenAI. Gpt-3.5-turbo system card. https://platform.openai.com/docs/models/gpt-3-5-turbo, 2023. 4

[30] OpenAI. Gpt-4v(ision) system card. https://openai.com/research/gpt-4v-system-card, 2023. 5, 6

[31] Viorica Pătrăucean, Lucas Smaira, Ankush Gupta, Adrià Recasens Continente, Larisa Markeeva, Dylan Banarse, Skanda Koppula, Joseph Heyward, Mateusz Malinowski, Yi Yang, Carl Doersch, Tatiana Matejovicova, Yury Sulsky, Antoine Miech, Alex Frechette, Hanna Klimczak, Raphael Koster, Junlin Zhang, Stephanie Winkler, Yusuf Aytar, Simon Osindero, Dima Damen, Andrew Zisserman, and João Carreira. Perception test: A diagnostic benchmark for multimodal video models. 2023. 5

[32] Shuhuai Ren, Linli Yao, Shicheng Li, Xu Sun, and Lu Hou. Timechat: A time-sensitive multimodal large language model for long video understanding. 2024. 3, 4

[33] Hao Shao, Shengju Qian, Han Xiao, Guanglu Song, Zhuofan Zong, Letian Wang, Yu Liu, and Hongsheng Li. Visual cot: Advancing multi-modal language models with a comprehensive dataset and benchmark for chain-of-thought reasoning. 2024. 2

[34] Dídac Surís, Sachit Menon, and Carl Vondrick. Vipergpt: Visual inference via python execution for reasoning. 2023. 1, 2

[35] Yi Wang, Kunchang Li, Xinhao Li, Jiashuo Yu, Yinan He, Chenting Wang, Guo Chen, Baoqi Pei, Rongkun Zheng, Jilan Xu, Zun Wang, et al. Internvideo2: Scaling video foundation models for multimodal video understanding. 2024. 3, 4

[36] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. 2022. 2

[37] Zejia Weng, Xitong Yang, Ang Li, Zuxuan Wu, and Yu-Gang Jiang. Open-vclip: Transforming clip to an open-vocabulary video model via interpolated weight optimization. 2023. 3, 4

[38] Bo Wu, Shoubin Yu, Zhenfang Chen, Joshua B Tenenbaum, and Chuang Gan. Star: A benchmark for situated reasoning in real-world videos. 2021. 2, 4, 5

[39] Junbin Xiao, Xindi Shang, Angela Yao, and Tat-Seng Chua. Next-qa: Next phase of question-answering to explaining temporal actions. 2021. 2, 4, 5

[40] Jihan Yang, Shusheng Yang, Anjali Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in Space: How Multimodal Large Language Models See, Remember and Recall Spaces. *arXiv preprint arXiv:2412.14171*, 2024. 2, 5, 6

[41] Zongxin Yang, Guikun Chen, Xiaodi Li, Wenguan Wang, and Yi Yang. Doraemongpt: Toward understanding dynamic scenes with large language models (exemplified as a video agent). 2024. 2

[42] Kexin Yi, Chuang Gan, Yunzhu Li, Pushmeet Kohli, Jiajun Wu, Antonio Torralba, and Joshua B. Tenenbaum. CLEVRER: collision events for video representation and reasoning. 2020. 2, 4, 5

[43] Shoubin Yu, Jaemin Cho, Prateek Yadav, and Mohit Bansal. Self-chained image-language model for video localization and question answering. 2023. 3, 4

[44] Zhou Yu, Dejing Xu, Jun Yu, Ting Yu, Zhou Zhao, Yueting Zhuang, and Dacheng Tao. Activitynet-qa: A dataset for understanding complex web videos via question answering. 2019. 5

[45] Zhou Yu, Lixiang Zheng, Zhou Zhao, Fei Wu, Jianping Fan, Kui Ren, and Jun Yu. Anetqa: A large-scale benchmark for fine-grained compositional reasoning over untrimmed videos. 2023. 2, 4, 5

[46] Hang Zhang, Xin Li, and Lidong Bing. Video-llama: An instruction-tuned audio-visual language model for video understanding. 2023. 5, 6

[47] Yuanhan Zhang, Bo Li, haotian Liu, Yong jae Lee, Liangke Gui, Di Fu, Jiashi Feng, Ziwei Liu, and Chunyuan Li. Llava-next: A strong zero-shot video understanding model. https://llava-vl.github.io/blog/2024-04-30-llava-next-video/, 2024. 1, 2, 3, 4, 6

[48] Zhuosheng Zhang, Aston Zhang, Mu Li, Hai Zhao, George Karypis, and Alex Smola. Multimodal chain-of-thought reasoning in language models. *arXiv preprint arXiv:2302.00923*, 2023. 2

[49] Orr Zohar, Xiaohan Wang, Yonatan Bitton, Idan Szpektor, and Serena Yeung-Levy. Video-star: Self-training enables video instruction tuning with any supervision. *arXiv preprint arXiv:2407.06189*, 2024. 2